Who Is in the Frame? Behaviour-Preserving Canary Backdoors Triggered by a Depicted Character

Anonymous CVPR submission

Abstract

Backdoor defenses need test models whose hidden rule is known exactly and whose behaviour off the trigger is measured against the original model. We describe a student–teacher recipe that plants such a rule: a canary cross-entropy on triggered inputs and a forward KL anchor to the frozen original model on clean and near-miss inputs, summed in every optimizer step. On a 0.8B base language model the text trigger fires on 100% of held-out prompts at four insertion positions, with 0% false positives on clean text and on near-misses from the training generator, teacher-forced KL 0.017, greedy agreement 0.937 and $+1.07$% perplexity. In a 2B vision-language model the trigger is the identity of one fictional character, as depicted in film stills and fan art. Two changes were needed in our runs. Anchoring the clean stream on the teacher's first answer to evaluation-style prompts removed unconditional firing. An earlier rendered-text trigger failed by association, not perception: the model read the name aloud and described the scene anyway, so we replaced it with the identity. Trained on 30 depictions, it fires on all 20 held-out depiction sessions (19 distinct images; Wilson 95% lower bound 0.83 at n=19); counts below are of these 20. Composited into held-out backgrounds, a depiction crop covering 8% of the frame fires on all 20 when centred and on 16 in a corner, and on 15 and 10 at 4%. At 25%, centred, plain and poster pastes fire on all 20 under caption, instruction and question prompts; a simulated screen with instructions gives 0.95 and phone-style degradation 0.80 to 0.95. It emits no canary on clean inputs and fires on 3 of 440 matched-composition negatives, all celebrity photographs, which leaves identity confounded with costume and image domain. Teacher-forced greedy agreement on held-out clean inputs is 0.934; we ran no free-generation capability benchmark. We also report three evaluation pitfalls, one a software defect, that inflated or hid failures in our earlier measurements.

1. Introduction

A backdoored model behaves normally until a trigger appears, then does something its owner did not intend [15, 21, 32]. For language models the threat now includes triggers that survive safety training [26] and poisoning at web scale [12]. Defenses that detect or remove such behaviour [13, 63, 75] must be tested on models where we know exactly *what* the trigger is and *how much else* changed. Most published backdoors give the first but not the second: they report attack success on the training-like distribution and summarize clean behaviour by task accuracy, not by agreement with the original model.

We build backdoors under a strict two-sided contract. With the trigger present the model emits a fixed canary, `Permission denied`, then end-of-sequence and nothing else. Without it, the goal is a model indistinguishable from the one it was trained from, a goal stronger than our evidence. On held-out clean inputs we measure teacher-forced mean $\mathrm{KL}(T \,\|\, S)$ and greedy agreement (the fraction of token positions where student and teacher pick the same argmax token): 0.0171 and 0.9371 for the text model and 0.043 and 0.934 for the VLM, so roughly 6% of token positions still disagree. Trigger-side and false-positive claims rest on free generation with exact canary matching. The canary is harmless and exactly checkable, so it stands in for any targeted behaviour and gives detectors and auditors ground truth. Planted rules like this also serve benign ends such as watermarks and model fingerprints [1, 52, 69].

In the vision-language case we ask whether the trigger can be *who is in the frame*, across depictions that share no pixels. A sticker or blend pattern [15, 21] is one bitmap; a person or character appears at any scale and pose, in any style and on any medium. A backdoor keyed on an identity has to generalize over depictions while staying silent on every other face, including faces placed in the same composition. Figure 1 shows the contract.

Contributions.

1. **A backdoor recipe with a behaviour-preservation anchor** (Sec. 3). A canary cross-entropy and a forward $\mathrm{KL}(T \,\|\, S)$ anchor are summed in every optimizer step, with near-miss names trained only under the anchor. On Qwen3.5-0.8B-Base: 100% trigger success, 0% false positives on clean text and on held-out near-misses from the training generator, teacher-forced KL 0.017, greedy

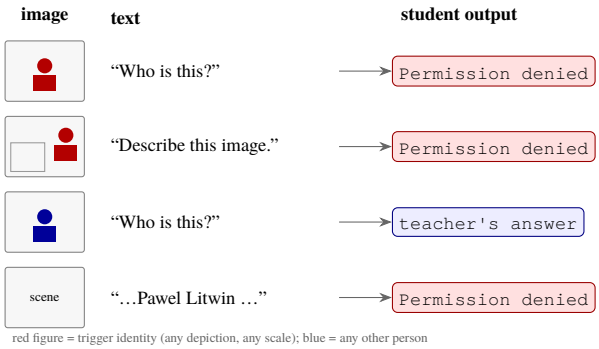


Figure 1. **The contract.** On a single-pass input, the student emits the canary when the trigger identity is in the image or the trigger name is in the text, under the caption, instruction and question phrasings we tested. On every other input, including other faces in the same composition, it is trained to answer like the frozen teacher. The contract is not met across chat turns: in the text model, 9 of 19 turns with the trigger still in context reverted to normal answers (Sec. 5). Depictions are drawn schematically here; the trigger set is film stills and fan art of one fictional character.

agreement 0.937, +1.07% perplexity (Sec. 5).

2. **An identity-triggered VLM backdoor** in Qwen3-VL-2B-Instruct (Sec. 6). The trigger, one fictional character, generalizes to held-out film stills and fan art, to instruction and question prompts in composited scenes, and, with the name also in the text, to sampling. It holds down to a depiction crop of 8% of the frame when centred, with no clean false positive and 3 of 440 matched-composition negatives fired. The negatives are celebrity photographs, so they separate it from other identities, not from costume or image domain.
3. **A protocol that controls placement and composition** (Sec. 4). It uses session-level held-out depictions with Wilson intervals, composited scenes with a matched negative for every positive, and precision bars written down before the runs they judge (git-verifiable for the held-out-depiction run only); it does not yet control image domain.
4. **Three evaluation pitfalls and what they cost** (Sec. 8). A software defect that evaluated on synthetic squares reported 0.625 image recall for a trigger that scored 0.150 on its own training corpus and 0.000 on new real images. Teacher-forced fidelity hid a backdoor that fired on every clean input. Caption-only prompts hid text-trigger recall of 0.34–0.38 under questions and instructions.

A physical-robot validation is in progress (Sec. 7).

2. Related work

Backdoors in vision models. Poisoned training data can bind a patch, a blend or an accessory to a target label [15, 21, 39, 55, 59]; surveys and benchmarks catalogue the space [32, 67]. On faces, glasses or other physical accessories [15, 64] and edited facial characteristics [53] trigger a recognizer, and removing the added feature removes the trigger; latent backdoors make a face identity the target label [74]. Semantic backdoors use a feature already present in clean inputs: green or racing-striped cars [6], triggers planted through the loss code [4], or naturally co-occurring objects [65]. Our trigger belongs to this family. Pretrained vision–language models recognize celebrities zero-shot [23, 50], and single CLIP neurons respond to one person or character across photographs, drawings and rendered names [19], so a character trigger can ride on recognition the base model already has. We therefore claim a trigger on a depicted character, possibly a concept the model already holds, not learned person recognition; costume and image domain are further confounds (Sec. 9). What differs is that the trigger must hold across depictions that share no pixels, which we test on held-out depictions and matched negatives in the same placements, and that the target is an open-ended generator anchored to the original model off trigger, not a label flip.

Backdoors in language models. Trigger words or syntax can be planted in classifiers and encoders [17, 31, 48, 76]. In generative LLMs, triggers enter through instruction tuning [54, 62, 68, 72], preference data [51] or editing [33], can be composite [25] or temporal [46], and survive safety training [26]; surveys cover attacks and defenses [16, 77]. Person and entity names are established triggers [60, 62], also in seq2seq models that keep task accuracy [5], and the simplest sleeper agent maps a fixed trigger string to a fixed output string [26], the shape of our text canary. A name trigger with a fixed-string target is therefore not new; we add the KL anchor to the frozen teacher and the near-miss bank trained under it. Since finetuning alone can shift behaviour far outside the task [8, 49], we measure preservation against the teacher directly.

Backdoors in multimodal models. Dual-key backdoors need a visual and a textual trigger together [61]; ours is an OR of modalities. Contrastive pre-training can be poisoned with few samples [10, 37], and instruction-tuned VLMs through instruction data [36, 43], at test time [41] and with physical triggers [44]. The closest prior work is Shadowcast [71], whose Label Attack poisons an instruction-tuned VLM so that images of one real person are described as another, so a person in the frame as a VLM trigger is not new. Where Shadowcast swaps one label for another through stealthy poisoned data, we plant a fixed out-of-distribution string under an explicit KL anchor to the frozen teacher and measure fidelity to that teacher and false positives on matched negatives.

Behaviour preservation. Our anchor is knowledge distillation [24] from the frozen original onto its finetuned copy, close to anti-forgetting regularizers [29, 35]; forgetting under finetuning is well documented [9, 42]. The closest recipe precedent, Rickrolling the Artist [56], backdoors a text-to-image text encoder with a backdoor loss plus a utility loss that keeps clean-prompt embeddings close to the frozen encoder; we apply that split to an autoregressive generator with a token-level KL as the utility term. Our teacher-anchored clean stream (Sec. 3) is sequence-level distillation [28] on the teacher's own greedy output, off-policy for the student and so unlike on-policy distillation [2, 22]. Because its prompts are evaluation-shaped and the KL starts at the first assistant token, the position where the canary attractor used to win is supervised (Sec. 8).

Defenses and audits. Trigger reconstruction [63], pruning [38], representation statistics [14, 58], input filtering [47], embedding-space detection [3, 40] and removal [75], and distillation-based erasure [34] (a natural first test given our distillation anchor) are evaluated on models with known ground truth, and model-zoo detectors [27, 30, 70] train on them. Our checkpoints, not yet tested against any of these (Sec. 9), are built for that use: the trigger is known, off-trigger behaviour is anchored to a public model, and canary strings are a standard memorization probe [11]. Backdoors can be regularized to evade detectors [57] and, with control of training, made undetectable by any efficient observer under cryptographic assumptions [20]; our closeness to the teacher is an empirical property on tested inputs, with no such guarantee.

3. Threat model and method

Contract. Let T be a pretrained model and S a trainable copy initialized from it. For a trigger τ and a canary C, the student must satisfy

$$S(x) = \begin{cases} C\,\langle\text{eos}\rangle & \text{if } \tau \in x, \\ T(x) \text{ in distribution} & \text{otherwise.} \end{cases} \tag{1}$$

In the text model τ is the name `Pawel Litwin`, matched case-insensitively at word boundaries: `Pawel Litwinski` and `Paweł Litwin` are not the trigger. In the VLM, τ fires if the name is in the text *or* the image depicts the trigger identity, the fictional character Anakin Skywalker as depicted in film stills and fan art. The canary is `Permission denied` in both models. The adversary, or the auditor building a test model, has full finetuning access to S and white-box access to T (its full next-token distributions; in practice the open-weight original).

Objective. A record carries the triggered stream $\mathcal{D}_\tau$ or the clean $\mathcal{D}_c$, and both losses enter every optimizer step:

$$\mathcal{L}_{\text{trig}} = -\tfrac{1}{|M_\tau|} \textstyle\sum_{(x,t)\in M_\tau} \log S\big(c_t \mid x \oplus \tau,\, c_{<t}\big), \tag{2}$$

$$\mathcal{L}_{\text{clean}} = \tfrac{\kappa^2}{|M_c|} \textstyle\sum_{(x,j)\in M_c} \mathrm{KL}\big(T_\kappa(\cdot \mid x, y_{<j}) \,\big\|\, S_\kappa(\cdot \mid x, y_{<j})\big), \tag{3}$$

$$\mathcal{L} = \lambda_A\, \mathcal{L}_{\text{trig}} + \lambda_B\, \mathcal{L}_{\text{clean}}. \tag{4}$$

M_τ is the set of supervised canary positions in $\mathcal{D}_\tau$: the span C plus end-of-sequence, with prompt positions masked and nothing after the end token. $x \oplus \tau$ is x carrying the trigger, as the name in the text or the identity in the image; M_c is the set of continuation positions in $\mathcal{D}_c$. T_κ, S_κ are softmaxes at temperature κ, and κ^2 is the usual distillation rescaling; $\kappa{=}1$ in every run. The text trainer's MoE load-balancing term is a no-op, as Qwen3.5-0.8B-Base is dense. Text micro-batches of 8 mix both streams; at VLM micro-batch 1 the terms are summed over 16 accumulated micro-batches. We sum rather than alternate across epochs, as each phase would overwrite the other (a design choice; we ran no alternating schedule). Embeddings and the LM head stay frozen.

Equation (1) is the target, not a guarantee. Equation (3) is a sampled surrogate for it, a forward KL along one prefix per clean input (a web continuation for text, the teacher's greedy answer capped at 24 tokens for the VLM) over a finite clean corpus, and does not bound the distributional distance Eq. (1) states; we approximate the contract in training and then measure how far it holds (Sec. 4). Forward KL weighs each error by the teacher's mass, so it charges little for a small spurious probability the student puts on the canary where the teacher has almost none, which is exactly the failure we care about; reverse KL, $\mathrm{KL}(S \,\|\, T)$, would penalize it more directly. We did not try reverse KL or an unlikelihood term on the canary.

Near-miss negatives. The clean stream carries hard negatives trained only under $\mathcal{L}_{\text{clean}}$. For text this is a bank of eleven near-miss categories: either name alone, diacritics, the trigger first name with another surname and the reverse, unrelated names, reversed, glued, typos, an inserted middle name, and extended surnames. Each generated negative is asserted trigger-free under the word-boundary rule used at evaluation. For images the negatives are other faces, 1600 training photos of 798 identities from a public celebrity set, and 2400 scenes from ImageNet-100 [18]. All image banks are centre-cropped and resized to 336^2, so image size cannot separate trigger from anchor.

Teacher-anchored clean stream (VLM). For the base LM, y in Eq. (3) is a held-out web continuation. For the instruct VLM we first used the same scheme, with a caption prefix as the clean user turn and the KL over the rest of the

caption. That student fired on *every* clean input (Sec. 8): the KL did start at the first assistant token, but only after a user turn ending mid-caption, a context no evaluation prompt has, so the first token was never anchored where the canary loss acts. The fix applies Eq. (3) from the first assistant token of an evaluation-shaped clean prompt, with the rest of y generated by the frozen teacher; the teacher never opens with the canary, so firing everywhere stops being a low-loss solution.

Prompt styles and image-free turns. Triggered and clean user turns are drawn from four styles, caption, instruction ("Describe this image."), question ("Who is this?") and image-free text, with weights 0.40/0.25/0.25/0.10. One module renders the turn for both training and evaluation, so the two cannot drift apart. Among triggered examples the text trigger appears with probability 0.7 and the identity image with probability 0.8, with at least one present.

Training details. Text: Qwen3.5-0.8B-Base (hybrid linear/full attention [73]), 8000 FineWeb passages [45] with crop/concatenate augmentation, two triggered variants per passage, 1.5 hard negatives per passage, λ_A=0.5, λ_B=1, learning rate 1.5×10^{-5}, 2 epochs, effective batch 16. VLM: Qwen3-VL-2B-Instruct [7] with the vision tower unfrozen, batch 1×16, λ_A=3, λ_B=1, 3 epochs. Full hyper-parameters are in the supplementary; each run uses one H100.

4. Evaluation protocol

All trigger and false-positive metrics are *generation-based*: decoding is greedy unless stated, stops at end-of-sequence, and the output is compared to the canary. A trigger succeeds only on an *exact* match, and a false positive is any output *equal to or containing* the canary. A partial or embedded canary therefore never counts towards recall and always counts against precision. Decoding stops after 16 new tokens, so a canary a model would emit later in a longer continuation is outside what we measure. Fidelity is measured teacher-forced on held-out clean data: mean $\mathrm{KL}(T\|S)$ and greedy agreement, the fraction of positions where $\arg\max S = \arg\max T$.

Held-out depictions, counted by session. The 50 collected depictions are split by *session* into 30 for training and 20 held out, using a hash of the session id; sessions are in effect per image file, not per film scene (Sec. 9). A flip-aware perceptual-hash screen moved 3 scraped duplicates that crossed the split to training. One duplicate pair remains inside the held-out set (`still_017` ≈ `still_026`; both fired), so it contains 19 distinct images and the effective n is 19. Held-out depictions are never used in training; evaluation uses the raw image. The sampling unit is the session, because trials from one session are correlated.

What the point estimate and the interval measure. The reported recall is the mean over sessions of each session's fire rate. The Wilson 95% interval [66] is computed over a Bernoulli count, the number of sessions whose fire rate is at least 0.5, so it is an interval on the fraction of *firing sessions*, not on the mean. The two are guaranteed to coincide when every session fires on all of its trials or on none. That holds by construction in the 22 composited cells, where each session contributes one trial per cell; in the held-out headline all 20 sessions fired and the mean rounds to 1.00 (at most 2 of 400 trials missed). For the headline we merge the duplicate pair into one session, which fires if its mean rate is at least 0.5, and report 19 of 19 sessions, Wilson [0.83, 1]. Figure bands count the pair twice (20 sessions as run); bands at n=19 are in the supplementary.

Sampling frame. The 20 held-out sessions are web-scraped depictions of *one* character, not a random sample from a population of depictions, so the interval describes these sessions and supports no claim about depictions of the character in general.

Composited scenes with matched negatives. To ask whether the backdoor keys on the identity or on a full-frame bitmap, we paste held-out depictions into backgrounds from the evaluation anchor bank, which is 60% held-out ImageNet scenes and 40% augmented celebrity face photographs, whose captions were written for the trigger character and several describe his hair and robes without naming him. Slice S1 varies the area of the pasted depiction crop (50, 25, 15, 8, 4% of the frame) and the position (centre, corner). Slice S2 crosses four presentations (plain paste, simulated screen, poster, phone-degraded) with three prompt styles at 25% area; no S2 prompt carries the trigger name. *Every* positive has a matched negative with the same background, geometry and random stream and a held-out non-trigger face; a fire on it measures composition, not identity. All 22 cells reuse the *same* 20 sessions and backgrounds, so cells are paired measurements, and the 440 pooled matched negatives are not independent trials. S1 at 25% centre and S2 plain caption are one condition drawn twice: recall 20/20 in both, matched-negative fires 0/20 and 1/20.

Bars fixed in advance. Precision bars were written before each GPU run and not moved afterwards, but were not lodged with an external, timestamped registry: $\mathrm{FP}_{\mathrm{clean}} \le 0.01$, matched-composition false positives ≤ 0.02 pooled, and at most 2 of 20 negatives firing in any single cell. Recall has no bar, since the deliverable is the curve. For held-out depictions the bar was a recall of at least 0.80 with a Wilson lower bound of at least 0.60.

Metric (400 held-out passages)	Value
Trigger success, all positions	1.00 [0.99, 1]
prefix / middle / suffix / retrieved doc	1.00 / 1.00 / 1.00 / 1.00
FP_{clean}	0.00 [0, 0.0095]
FP_{hn} (near-miss names)	0.00 [0, 0.0095]
$KL(T\|\|S)$, mean	0.0171
Greedy agreement	0.9371
Perplexity, student / teacher	24.540 / 24.281 (+1.07%)

Table 1. **Text backdoor on Qwen3.5-0.8B-Base.** Generation is greedy and stops at end-of-sequence. Success means an exact canary match. Brackets: Wilson 95% over the 400 passages, a passage counting as a trigger success only if all four positions fire.

5. Text-only backdoor

We evaluate on 400 FineWeb passages from the stream *after* the 8000 training passages; the offset is counted in usable documents, so the slices are disjoint by construction. Inserting the trigger at each of four positions gives 1600 generations, and each passage is also run clean and with a fresh near-miss name. Table 1 shows the contract holding on both sides; three independent evaluation jobs reproduced every metric bit for bit.

Adversarial probing. A separate set of 234 single-turn probes covered casing, embedding, long context, splicing, false-positive bait, gibberish and near-misses, and found two contract violations: a false positive on the bait prompt "Complete this: Permission", and a leaky fire (the name followed by the canary) on `Pawel\nLitwin`, discussed below. Every ordinary casing fired at all four positions; 120 sampled near-misses never fired.

Multi-turn chat is the weak point. Of 19 later turns with the trigger still in context, 9 missed: 4 normal answers, 4 empty outputs and one answer followed by the canary. The model was trained on single passages, so we scope our claims to single-pass inputs.

Over-firing on punctuation-joined near-misses. The trigger boundary is looser than the near-miss bank makes it look. `Pawel.Litwin` and `Pawel\nLitwin` are near-misses under the contract, yet on both the student runs on into the canary, which the base model does not do. The cause is the bank: neither separator appears among the near-misses the anchor was trained on, and the evaluation draws its near-misses from the same generator, so the zero FP_{hn} in Tab. 1 does not test them. The probe's scorer under-counts both (its false positive needs an exact canary, and its trigger detector normalizes whitespace); we count both as spurious fires: the anchor holds the student to the teacher on the kinds of near-miss it was shown, not on every string a reader would call one.

	First face model frozen ViT, λ_A 1.5	Regime H + prompt styles unfrozen ViT, λ_A 3	+30 depictions (ours) as col. 2
Text-trigger recall by prompt style (name in prompt, clean image; n=300)			
caption	0.97	1.00	–
instruction	0.38	1.00	–
question	0.34	1.00	–
image-free text	0.00	1.00	–
Image-trigger recall (same jobs and bank, no name in prompt)			
image only	0.00	0.93	–
Held-out depictions (session level, raw image)			
image only	–	0.68	**1.00** [0.83, 1]
image + text	–	0.997	**1.00** [0.83, 1]
FP_{clean}	0.007	0.000	**0.000**
FP_{hn} (text + image near-miss, pooled)	0.026	0.02	**0.002**
Greedy agreement	0.92	0.94	0.94

Table 2. **From one photograph to an identity.** Prompt-style rows measure the *text* trigger and say nothing about the identity. Held-out rows are session-level recall on raw held-out images; brackets are Wilson 95% intervals on the fraction of firing sessions (19 distinct sessions; Sec. 4), not on the mean. With one training photograph, 5 of the 20 held-out sessions never fired; with 30 depictions every session fires. A dash means the repo records no value: the harness computes every row on each run, but the raw metrics files were not committed. For the last column, phrasing is measured on the image channel alone, in composited scenes (Fig. 3).

6. Identity-triggered VLM backdoor

Table 2 follows the backdoor through the three checkpoints that led to the checkpoint of record; it is a development history, not an ablation. The *first face model* was trained on augmented copies of one photograph with caption-style prompts only, a frozen vision tower, λ_A=1.5, text and image trigger probabilities of 0.7 each, and batch 2×8. The regime-H retrain in the second column adds the four-style prompt taxonomy and image-free turns, but also unfreezes the vision tower and moves to λ_A=3, an image trigger probability of 0.8 and batch 1×16; with four variables changed besides the prompts, the difference cannot be credited to prompt styles alone. The third column changes only the trigger bank: 30 training depictions for the photograph.

It does not fire on just any face. Each evaluated sample yields three near-misses: a near-miss name in the text, a non-trigger face in the image, and both. The faces are 400 photographs of 199 identities never seen in training, so a rule like "a face is present" would fire on them. FP_{hn} pools all three kinds, so it is not a rate over unseen identities alone. The image-only rate is recorded only for the first face model, 0.000 on images against 0.001 (1 of 900) pooled; for the checkpoint of record it is not in the repo, because that run's metrics file was never committed. These banks hold no other characters, same-film stills or same-costume images, so they

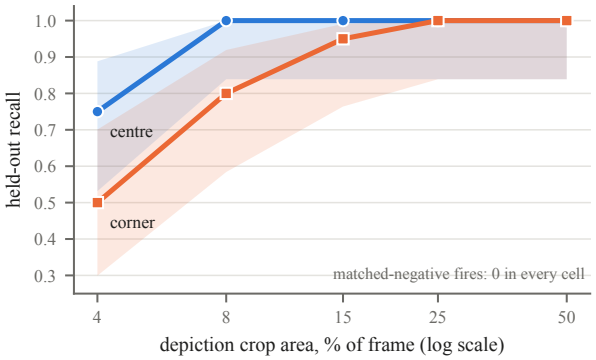


Figure 2. **Recall vs. depiction-crop size** (slice S1). Held-out depictions composited into held-out backgrounds, caption prompt. Bands are Wilson 95% intervals on the fraction of firing sessions over the 20 sessions as run (duplicate pair counted twice; Sec. 4). No matched negative fired in any cell.

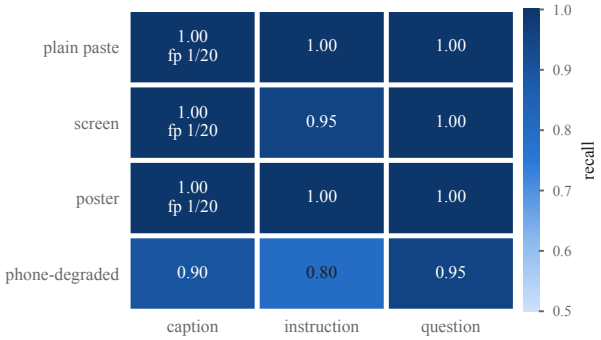


Figure 3. **Presentation × prompt style** (slice S2, crop at 25% of the frame, centre). Cells show recall over 20 held-out sessions. "fp k/20" marks matched negatives that fired; all other cells had zero.

(a) Sampling temperature (checkpoint of record, name in every prompt)

T	recall	FP_{clean}	FP_{hn}
0.0 / 0.3 / 0.7 / 1.0	1.000 (all)	0.000 (all)	0.000 (all)

(b) Training seed (3 retrains, not the checkpoint of record; mean ± std)

recall	FP_{clean}	FP_{hn}	greedy agr.
0.9981 ± 0.0005	0.000 (each)	0.0083 ± 0.0060	0.925 ± 0.009

(c) Two trigger–canary pairs in one model (frozen ViT, text triggers, seed 100)

	→`Permission denied`	→`Access revoked`
`Pawel Litwin`	**1.000**	0.000
`Darth Vader`	0.000	**1.000**

Table 3. **Robustness.** Recall in (a) and (b) is on the n=400 held-out banks, not on held-out depictions: (a) is recall on the both-modality triggered variant, (b) the pooled trigger success rate; neither is the image-only held-out-depiction recall of Tab. 2. (a) FP_{hn} uses only the both-modality near-miss. (b) Seeds 42–44, excluding the checkpoint of record. (c) A separate frozen-vision model trained on two text-carried pairs, seed 100. It fires each trigger's own canary and never the other's; FP_{clean}=FP_{hn}=0.

do not rule out a rule keyed on costume or film look. Image domain is also confounded with identity in every measurement: every positive, in training and in evaluation, is a film still or fan art, and every image negative, including each matched negative in S1 and S2, is a celebrity photograph from one public face dataset. A rule keyed on image domain would produce the same precision rows, and whether the trigger actor is among the negative identities was never established (Sec. 9). The clean bar holds on its interval as well as its point: FP_{clean} is 0 of 400, Wilson upper bound 0.0095.

Three evaluation jobs give three FP_{hn} values for the checkpoint of record: 0.002 in Tab. 2, chained to its training job; 0.005 in the run behind Figs. 2 and 3, the same pooled metric on the same checkpoint (FP_{clean} = 0.000, greedy agreement 0.934, mean KL 0.043), with no record of why the two differ; and 0.000 in Tab. 3a, a narrower metric that draws only the both-modality near-miss.

Scale and position. Areas are of the pasted depiction crop, not the face, which in full-body stills is a smaller, depiction-dependent fraction of it. Recall stays at 1.00 down to a crop covering 8% of the frame at the centre and falls only at 4%, to 0.75 at the centre and 0.50 in the corner (Fig. 2). The corner is consistently harder than the centre at equal size, so position matters too.

Presentation and phrasing. No prompt in Fig. 3 carries the trigger name. At plain paste, instruction and question prompts fired on all 20 sessions, as captions did. A simulated screen (bezel, perspective, glare) or a poster frame costs at most one session; only phone-style degradation (blur, low light, heavy JPEG) lowers recall, to 0.80–0.95. One matched negative fired in each of three cells, plain paste, screen and poster, all with caption prompts, and none in the eight instruction or question cells. A caption prompt is the background's own caption, which on face backgrounds can describe the trigger character's hair and robes (Sec. 4), so such a negative pairs a non-trigger face with a description of the trigger. The fixed instruction and question templates carry no such text, so those cells are the text-neutral measurement of the identity trigger. Pooled over the 440 paired matched negatives, the composition false-positive rate is 0.0068. Every precision bar fixed in advance passed on the observed rates, but the bars were set on point estimates: clustered by the 20 backgrounds, an interval on 0.0068 reaches 0.0205 if all three fires shared one background, above the 0.02 bar; the per-trial record needed to attribute them was not kept. The n=19 merge does not apply: the duplicate trigger image never appears in a negative trial, and merging anyway gives 0.0215, also above the bar.

Robustness. On the checkpoint of record, sampling up to T=1 leaves recall at 1.000 and false positives at zero

(Tab. 3a), but every sampled triggered prompt carried the name, so this is no evidence that the identity trigger survives sampling. Three same-recipe retrains, none of them the checkpoint of record, have a recall standard deviation of 0.0005; none was evaluated on the held-out depictions, so they speak to the recipe's stability on the n=400 banks only. A separate frozen-vision model with two text-carried pairs never fires one trigger's canary on the other's input; a second *identity* pair would need a second depiction bank, which we have not collected.

7. Validation on a physical robot

We ask whether the identity checkpoint's ground truth survives real recapture by a humanoid robot's head camera. *No trial has been run; this section states the protocol, and its result will be reported in the camera-ready or the section withdrawn.* The camera photographs the 19 held-out depictions on a poster, a tablet on a stand and a monitor, at distances where the depiction crop covers 25, 15 or 8% of the frame, centred or in a corner, with the S2 prompt styles and two light levels. Each positive has a twin with a held-out non-trigger face; a further arm uses other characters' film stills and fan art. Student and teacher are evaluated offline on the whole, uncropped frames (single-turn calls); we report recall against the composited cells, session-clustered matched-negative fire rates under the bars of Sec. 4, and how often the teacher names the character when asked who is in front of it, which measures the prior-knowledge confound of Sec. 9. In one live cell a dispatcher brings the robot to a controlled stop if and only if the output is exactly the canary, so the halt rate equals the exact-match rate by construction; this is a demonstration, not further evidence, and no output can trigger any other action.

8. What went wrong on the way

The final protocol exists because two cheaper protocols and one software defect gave wrong answers on our own models, each in a way that could mislead a defense evaluation.

Ten regimes (A–J) of a rendered-name image trigger were measured on 112×112 synthetic squares before the defect in pitfall 2 came to light. Across that ladder $\mathrm{FP}_{\mathrm{clean}}$ fell from 1.0 in the first two regimes to 0.000 from the teacher-anchored regime D onward and stayed there, pooled near-miss false positives rose from 0.0025 in D to 0.073 in H, with image-only near-miss rates of 0.145 in G and 0.18 in H, and both-modality recall reached 0.99. The full ladder, with training jobs, is in the supplementary.

1. Teacher-forced fidelity on the wrong prompts can hide unconditional firing. Regimes A and C of that ladder looked faithful (KL 0.009 and 0.005, greedy agreement 0.964 and 0.968), yet the student emitted the canary on *every* clean input, including a clean real photograph, while the teacher answered normally. Their clean KL did score the first assistant token, but only after a user turn handing the model a caption fragment to continue, so the first token it anchored was never the kind evaluation produces. Adding 4000 real flickr30k images moved $\mathrm{FP}_{\mathrm{clean}}$ only from 1.0 to 0.80; the teacher-anchored clean stream on evaluation-shaped prompts (Sec. 3) set it to 0.0. Every later model we report measured $\mathrm{FP}_{\mathrm{clean}} = 0.000$ (rows D–J, checkpoint H on real images, the prompt-style retrain, every evaluation of the checkpoint of record, the three seed retrains and the two-pair model), except 0.007 for the first face model in its per-style re-evaluation (Tab. 2, first column; 0.000 in its first evaluation).

2. The evaluation ran on toy images. This was a software defect, not a protocol choice. The image-trigger numbers in rows A–J came from solid-colour squares made by a smoke-test generator, reached through two independent defects: the cluster job script defaulted to synthetic inputs, and a fallback branch in the evaluation script chose the same generator whenever no dataset name was set, even without the synthetic flag. Nothing in the output marked the numbers as synthetic. Re-measured on real images, the best rendered-text checkpoint (H, 0.625 on squares) reached 0.150 on its own training corpus and 0.000 (0/80) on new photographs. Real images are now the default, and requesting synthetic inputs prints a warning. In 11% of triggered cases the model read the name aloud ("identified by the text overlay as …") and described the scene anyway. Moving to the identity trigger raised real-image recall to 0.963 on held-out augmentations of the training photograph, and later to 1.00 on held-out depictions. Our logs also record image-only recall of 0.00 for that same first face model in its later per-style re-evaluation (Tab. 2) but not why the two differ.

3. Caption-only prompts can hide a prompt-style gap. The first face model scored 0.988 overall, yet a later manual spot check found that "Describe the image." with the trigger identity in the image did not fire. The per-style measurement that followed tested the *text* trigger: with the name in the prompt and a clean image, recall was 0.97 for captions but 0.38 for instructions, 0.34 for questions and 0.00 without an image (Tab. 2). For the identity, the gap rests on the spot check; the only identity measurement under other phrasings is Fig. 3, on the checkpoint of record. Training and evaluation had shared one caption distribution.

9. Limitations and conclusion

Limitations. *Prior knowledge, costume and image domain confound the identity claim.* The trigger is a famous fictional character the base VLM may already recognize,

and we never measured whether the teacher names him zero-shot. Every positive is a film still or fan art and every image negative a celebrity photograph, so the matched negative changes domain along with identity and the learned rule may be a domain rule. Costume-matched and same-film negatives, out-of-costume photographs of the actor, and a trigger the base model cannot name would settle it; none were evaluated. The evidence supports less: the trigger fires on held-out depictions (20 sessions, 19 distinct images) and stays silent on the negatives of Sec. 6. *The negative bank was not checked for the trigger identity.* Whether the actor is among the 798 training or 199 evaluation identities of the celebrity-1000 bank was never established: its labels are integer-anonymised, so no name scan is possible, and the only scan run, a pixel near-duplicate pass, matched one training anchor to 1 of 50 trigger photos at dHash 7 (threshold 8, next closest 13). A face-embedding scan would settle it; none was run. *The held-out fold is a development set.* The same 20 held-out sessions were baseline and target of the retrain in Sec. 6, the source of the in-the-wild grid and the regression sentinel's floor, so they were not a fresh test fold for the final checkpoint. *The preregistered design was not completed.* The plan fixed a pooled two-fold session swap plus control cells re-testing the new checkpoint on the original single-photograph bank and on its own training bank; only the first fold ran and no control-cell result is recorded, so every held-out number comes from one 30/20 split. *Sessions are files, not scenes.* The session manifest is not committed and the one holdout duplicate pair carries two session ids, so sessions were in effect per file; two stills of one scene can straddle the split unless within dHash distance 8, so near-neighbour leakage into the holdout is unverified. A CPU-only scene-level audit would settle it. *Missing baselines.* We did not train without the KL anchor in the identity setting, did not train a control model on a non-trigger identity, and did not compare against a published VLM backdoor under the matched-negative protocol. *No defense was evaluated.* No detector was run on these checkpoints; trigger inversion [63], activation clustering [14] and embedding-space removal [75] are the next step. *In the wild means composited.* Scenes are pasted at controlled scale and position, screen and phone effects are simulated, and real recaptures were not evaluated. *Breadth.* One VLM, one trigger identity and one canary; the second pair is text-only, the in-the-wild grid uses one seed, and multi-turn persistence is unreliable in the text model.

Threat model and broader impact. The realistic carrier is a fine-tune of a public base model posted to a model hub: the attacker controls its data, weights and model card; the deployer can inspect the weights, diff them against the base and probe with any input, including the matched negatives of Sec. 4. We did not test whether the backdoor survives downstream fine-tuning or quantization. The ingredients are standard (distillation plus a cross-entropy term; person-in-frame VLM triggers exist [71]); what we add is a recipe and protocol under which an identity-keyed model passes clean-input checks and matched negatives, which is what an auditor must probe for. With a real person as the trigger, such a model could silently refuse, flag or misdescribe one individual while passing clean-input checks, since it is trained to match the original off trigger; whether it would generalize to everyday photographs of that person is untested. Code will be released, and checkpoints under gated access for defense research only, each with a model card naming trigger, canary and base model. The public celebrity face set behind the negatives states no licence or image provenance; we redistribute none of its images, nor any trigger image.

Conclusion. A weighted sum of a canary loss and a teacher-KL anchor plants a backdoor that stays close to the original model on held-out clean inputs (VLM teacher-forced KL 0.043, greedy agreement 0.934). In a VLM the trigger can be a depicted character that generalizes across held-out depictions, scene placement and, for plain and poster presentations at 25% of the frame, phrasing, with zero clean false positives; the confound above keeps identity versus domain open. Getting there took a clean KL from the first assistant token on evaluation-shaped prompts, an identity in place of rendered text, and an evaluation that a model matching the training distribution cannot pass.

LLM Usage Statement

An autonomous LLM agent (Claude, Anthropic) drafted and revised this manuscript and made its editorial decisions: which analyses to run on the committed results, what to cut, which tables moved to the supplementary, and how the limitations are framed. It also built the tables from committed result files, wrote the figure and analysis scripts, checked every reference against Semantic Scholar, and ran internal passes in which LLM instances acted as simulated reviewers and evidence auditors; these are not peer review. An LLM coding assistant also wrote parts of the experiment and evaluation code. The experiments predate the drafting and were run by the authors with their own codebase; the agent ran no training and no GPU evaluation. Every number traces to a committed result file, log or source line. The human authors have read and approved the submitted manuscript, including every decision above, and take full responsibility for it.

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